

4.

Image Restoration.

4.1. Introduction:

- Image Restoration is the class of methods that aims to remove or reduce the degradations that have occurred while the digital image was being obtained.
- The objective of image restoration is to improve a given image in some predefined sense.
- Image Enhancement is a subjective process whereas image restoration is an objective process.

4.2. A model of the Degradation/Restoration Process:

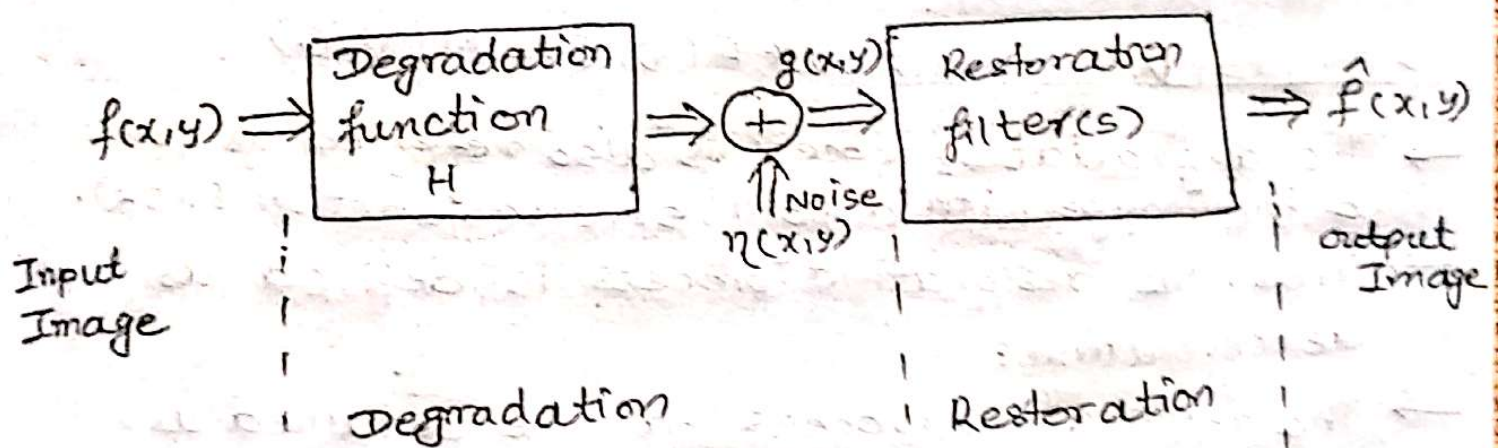


Fig. A model of the image Degradation/Restoration process

→ If $f(x,y)$ and $\eta(x,y)$ be the input image and additive noise term respectively then, the degraded image $g(x,y)$ is given by

$$g(x,y) = H[f(x,y)] + \eta(x,y)$$

→ The degraded image can be defined in spatial domain as,

$$g(x,y) = h(x,y) * f(x,y) + \eta(x,y)$$

where $h(x,y)$ = spatial representation of degradation function

→ The degraded image in spatial domain is equivalent to the convolution frequency domain and is represented as,

$$G(u,v) = H(u,v) * F(u,v) + N(u,v)$$

→ The degradation process is also referred as convolution of the image with a point spread function (PSF).

→ Thus, the restoration process is defined as deconvolution.

→ The restoration process can be obtained as,

$$\hat{f}(x,y) = R[g(x,y)], \text{ in linear spatial invariant form}$$

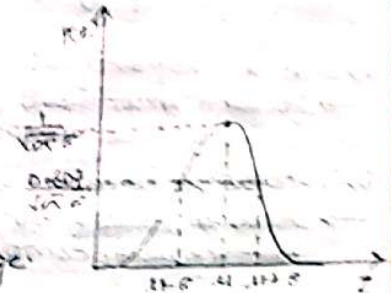
$$\hat{f}(u,v) = R[G(u,v)], \text{ in deconvolution form}$$

4.8. Noise Models in Image with PDF:

i) Gaussian noise:

→ Gaussian noise models are simple to consider and the PDF of Gaussian random variable (z) is given as,

$$P(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(z-\mu)^2/2\sigma^2}$$



a) Spatial Properties:

→ It appears as random variations with a smooth & continued distribution across image.

b) Frequency Properties:

→ It also affects both low and high frequency components.

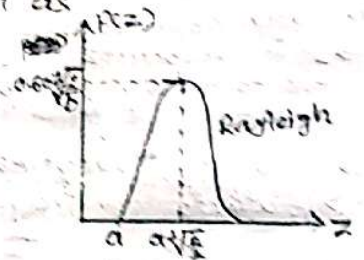
c) Filtering Technique:

→ Gaussian Smoothing or low pass filtering is effective for reducing Gaussian noise.

ii) Rayleigh noise:

→ Unlike Gaussian distribution, the Rayleigh distribution is not symmetric. The PDF is given as

$$P(z) = \begin{cases} \frac{z}{\sigma^2} e^{-z^2/2\sigma^2} & ; z \geq 0 \\ 0 & ; z < 0 \end{cases}$$



a) Spatial Properties:

→ It manifests as amplitude variation in the image.

b) Frequency Properties:

→ It affects both low & high frequency components of an image.

c) Filtering Technique:

→ It uses Adaptive filters for reducing Rayleigh noise.

iv) Gamma Noise (Erlang Noise):
 → The PDF of Erlang (Gamma) Noise is given by

$$p(z) = \begin{cases} \frac{a^b z^{b-1}}{(b-1)!} e^{-az} & ; \text{for } z \geq 0 \\ 0 & ; \text{for } z < 0 \end{cases}$$

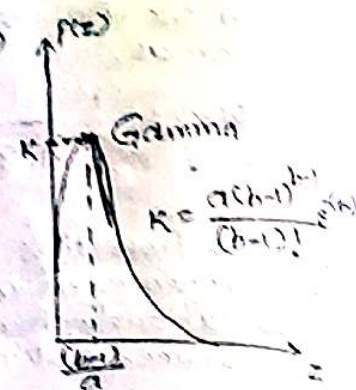
α) Spatial Properties:

→ It can manifest as variation in amplitude across image.

β) Frequency Properties:

→ It also affects both low pass & high pass frequency components.

γ) Filtering Technique:



v) Exponential Noise:

→ The PDF of exponential noise is given as:

$$p_z(z) = \begin{cases} ae^{-az} & ; z \geq 0 \\ 0 & ; z < 0 \end{cases}$$

α) Spatial Properties:

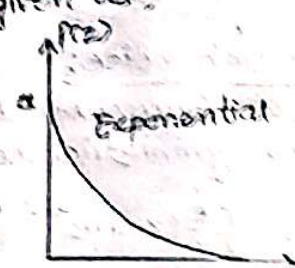
→ It is associated with random events occurring independently over time.

β) Frequency Properties:

→ It affects low frequency components of image or signal.

γ) Filtering Technique:

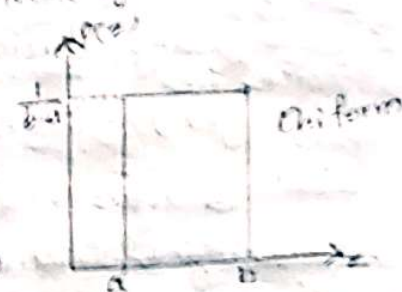
→ It uses low pass filtering for reducing noise.



vi) Uniform noise:

→ The PDF of uniform noise is given by

$$p(z) = \begin{cases} \frac{1}{b-a} & ; a \leq z \leq b \\ 0 & ; \text{otherwise} \end{cases}$$



α) Spatial Properties:

→ It manifests as constant value added to each pixel's intensity in image.

β) Frequency Properties:

→ It primarily affects low frequency components of an image.

γ) Filtering Technique:

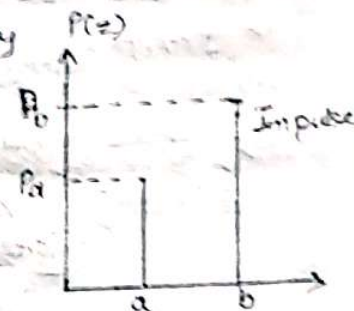
→ It uses mean filtering technique for reducing noise.

vii) Salt and Pepper (Impulse) Noise:

→ The PDF of impulse noise is given by

$$p(z) = \begin{cases} p_a & ; \text{for } z=a \\ p_b & ; \text{for } z=b \\ 0 & ; \text{otherwise} \end{cases}$$

→ If is a, gray level b will appear as a dot in image.



α) Spatial Filter Properties:

→ It appears as isolated randomly occurring bright and dark pixels.

β) Frequency Properties:

→ It also affects both low & high frequency components.

γ) Filtering Technique:

→

Level 8:

- If $Z_{min} < Z_{xy} < Z_{max}$, output Z_{xy}
Else output Z_{med}

where

Z_{min} : Min. Intensity value in S_{xy}

Z_{max} : Max. Intensity value in S_{xy}

Z_{med} : Median Intensity value in S_{xy}

Z_{xy} = Intensity value at co-ordinate (x, y)

S_{xy} = maximum allowed size of the adaptive filter window

→ It has three purposes:-

i) Remove Impulse Noise

ii) Avoid Smoothing of other Noise

iii) Reduce distortion

4.5. Periodic Noise Reduction by Frequency Domain Filtering.

→ Periodic noise is an image that arises typically from electrical or electromechanical interference during image acquisition.

→ Periodic noise can be reduced typically significantly via frequency domain filtering.

→ Periodic noise reduction technique uses a selective filter to isolate the noise.

→ There are 5 types of selective filters used for periodic noise reduction.

1. Band Reject Filter:

→ Band reject filter is used for removing periodic noise from an image involves removing a particular range of frequencies from that image.

→ Types of Band reject filters:

a) Ideal Band Reject Filter:

→ An ideal band reject filter is given as -

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) < D_0 - \frac{W}{2} \\ 0 & \text{if } D_0 - \frac{W}{2} \leq D(u, v) \leq D_0 + \frac{W}{2} \\ 1 & \text{if } D(u, v) > D_0 + \frac{W}{2} \end{cases}$$

b) Butterworth Band Reject filter:

→ A butterworth band reject filter is given as,

$$H(u, v) = \frac{1}{1 + \left[\frac{D(u, v)W}{D_0^2 - D_0^2} \right]^{2n}}$$

iii) Gamma Noise (Erlang Noise):
 → The PDF of Erlang (Gamma) Noise is given by

$$p(z) = \begin{cases} \frac{a^b z^{b-1}}{(b-1)!} e^{-az} & ; \text{for } z \geq 0 \\ 0 & ; \text{for } z < 0 \end{cases}$$

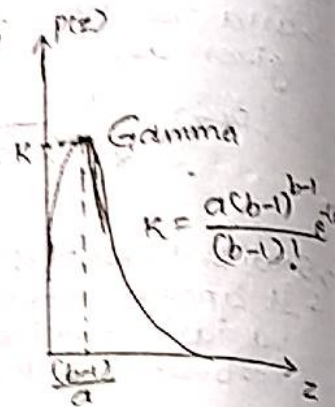
a) Spatial Properties:

→ It can manifest as variation in amplitude across image.

b) Frequency Properties:

→ It also affects both low pass & high pass frequency components.

c) Filtering Technique:



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→ The PDF of exponential noise is given as:

$$p(z) = \begin{cases} ae^{-az} & ; z \geq 0 \\ 0 & ; z < 0 \end{cases}$$

a) Spatial Properties:

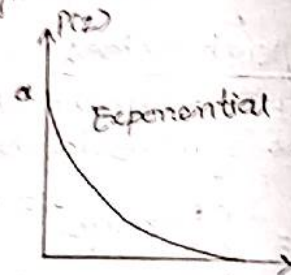
→ It is associated with random events occurring independently over time.

b) Frequency Properties:

→ It affects low frequency components of image or signal.

c) Filtering Technique:

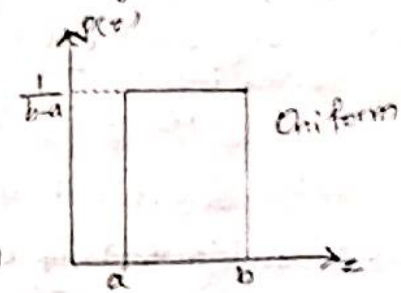
→ It uses low pass filtering for reducing noise.



v) Uniform noise:

→ The PDF of uniform noise is given by

$$p(z) = \begin{cases} \frac{1}{b-a} & ; a \leq z \leq b \\ 0 & ; \text{otherwise} \end{cases}$$



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→ It manifests as constant value added to each pixel's intensity in image.

b) Frequency Properties:

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c) Filtering Technique:

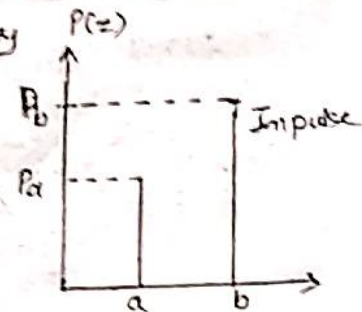
→ It uses mean filtering technique for reducing noise.

vi) Salt and Pepper (Impulse) Noise:

→ The PDF of impulse noise is given by

$$p(z) = \begin{cases} p_a & ; \text{for } z=a \\ p_b & ; \text{for } z=b \\ 0 & ; \text{otherwise} \end{cases}$$

→ If $b > a$, gray level b will appear as a dot in image.



a) Spatial Properties:

→ It appears as isolated, randomly occurring bright and dark pixels.

b) Frequency Properties:

→ It also affects both low & high frequency components.

c) Filtering Technique:

→

4.4. Noise Models Restoration in the presence of Noise

Only Spatial Filtering:

→ The only degradation present in an image is noise, i.e.,

$$g(x,y) = f(x,y) + \eta(x,y)$$

$$\text{or, } G(u,v) = F(u,v) + N(u,v)$$

→ Following techniques can be used to reduce the noise effect:

i) Spatial Noise Filters

ii) Adaptive Spatial Filters

4.4.1. Spatial Noise Filters:

→ In spatial noise filters, image independently varies from one location to another.

→ Arithmetic mean:

$$\hat{f}(x,y) = \frac{1}{mn} \sum_{(s,t) \in S_{xy}} g(s,t)$$

ii) Geometric mean:

$$\hat{f}(x,y) = \left[\prod_{(s,t) \in S_{xy}} g(s,t) \right]^{1/mn}$$

iii) Harmonic mean:

$$\hat{f}(x,y) = \frac{mn}{\sum_{(s,t) \in S_{xy}} \frac{1}{g(s,t)}}$$

iv) Median:

$$\hat{f}(x,y) = \text{median}_{(s,t) \in S_{xy}} (g(s,t))$$

i) Max:

$$\hat{f}(x,y) = \max_{(s,t) \in S_{xy}} \{g(s,t)\}$$

ii) Min:

$$\hat{f}(x,y) = \min_{(s,t) \in S_{xy}} \{g(s,t)\}$$

where, m and n = variables denote image rows & columns spanned by the filter, respectively.

S_{xy} = denotes an $m \times n$ sub image of the input noisy image g .

$\hat{f}(x,y)$ = filter response at (x,y) condition (coordinates)

(x,y) = coordinates of sub image

4.4.2. Adaptive Spatial Filters:

→ An adaptive spatial filter is one whose behavior can change based on statistical characteristics of an area within image.

→ An adaptive expression can be written as:

$$\hat{f}(x,y) = g(x,y) - \frac{\sigma_n^2}{\sigma_f^2} [g(x,y) - m_c]$$

→ The adaptive mean filtering algorithm uses too processing levels for adaptive spatial filters that can be denoted as:

i) Level A:

- If $Z_{min} < Z_{med} < Z_{max}$ → go to level B

Else, increase the window size

- If $Z_{window size} \leq S_{max}$, repeat level A

Else output Z_{med} .

Level B:

- If $Z_{min} < Z_{xy} < Z_{max}$, output Z_{xy}
Else output Z_{med}

where,

Z_{min} = Min. Intensity value in S_{xy}

Z_{max} = Max. Intensity value in S_{xy}

Z_{med} = Median Intensity value in S_{xy}

Z_{xy} = Intensity value at co-ordinate (x, y)

S_{xy} = minimum allowed size of the adaptive filter window

→ It has three purposes:-

i) Remove Impulse Noise

ii) Provide Smoothing of other Noise

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4.5. Periodic Noise Reduction by Frequency Domain Filtering:

→ Periodic noise is an image that arises typically from electrical or electromechanical interference during image acquisition.

→ Periodic noise can be reduced typically significantly via frequency domain filtering.

→ Periodic noise reduction technique uses a selective filter to isolate the noise.

→ There are 3 types of selective filters used for periodic noise reduction:-

1. Band Reject Filter:

→ Band reject filter is used for removing periodic noise from an image, involves removing a particular range of frequencies from that image.

→ Types of Band reject filters:-

a) Ideal Band Reject Filter:

→ An ideal band reject filter is given as:-

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) < D_0 - \frac{W}{2} \\ 0 & \text{if } D_0 - \frac{W}{2} \leq D(u, v) \leq D_0 + \frac{W}{2} \\ 1 & \text{if } D(u, v) > D_0 + \frac{W}{2} \end{cases}$$

where

b) Butterworth Band Reject filter:

→ A butterworth band reject filter is given as,

$$H(u, v) = \frac{1}{1 + \left[\frac{D(u, v)W}{D_0^2 - D_0^2} \right]^{2n}}$$

§ Gaussian Band Reject Filter:

→ A Gaussian band reject filter is given by

$$H(u, v) = 1 - e^{-\frac{1}{2} \left[\frac{D^2(u, v) - D_0^2}{D(u, v) W} \right]^2}$$

2. Band Pass Filter:

→ A band pass filter performs the opposite operation of band reject filters.

→ A band pass filter allows a specific frequency band of the image to be passed and blocks the rest of frequencies.

→ The transfer function $H_{bp}(u, v)$ of a band pass filter can be obtained from a corresponding band reject filter with transfer function $H_{br}(u, v)$ by using this equation,

$$\text{i.e. } H_{bp}(u, v) = 1 - H_{br}(u, v)$$

→ These filters cannot be applied directly on an image because it may remove too much details of an image.

→ These are effective in isolating the effect of an image of selected frequency bands.

3. Notch Filters:

→ A notch filter rejects (or passes) frequencies in predefined neighborhoods about a center frequency.

→ Due to symmetry of the Fourier transform notch filters must appear in symmetric pairs about the origin.

→ The transfer function of an ideal notch filter of radius D_0 with centers $a(u_0, v_0)$ and by symmetry at $(-u_0, -v_0)$ is

$$D_1(u, v) = \sqrt{(u - M/2 - u_0)^2 + (v - N/2 - v_0)^2}$$

$$D_2(u, v) = \sqrt{(u - M/2 + u_0)^2 + (v - N/2 + v_0)^2}$$

i) Ideal Notch Filter:

$$H(u, v) = \begin{cases} 0 & \text{if } D_1(u, v) \leq D_0 \text{ or } D_2(u, v) \leq D_0 \\ 1 & \text{otherwise} \end{cases}$$

$$D_1(u, v) = \sqrt{(u - M/2 - u_0)^2 + (v - N/2 - v_0)^2}$$

ii) Butterworth Notch Filter:

$$H(u, v) = \frac{1}{1 + \left[\frac{D_0^2}{D_1(u, v) D_2(u, v)} \right]^{1/2}}$$

iii) Gaussian Notch Filter:

$$H(u, v) = 1 - e^{-\frac{1}{2} \left[\frac{D_1(u, v) D_2(u, v)}{D_0^2} \right]}$$

Q. What are the main source of noise in image.
Compare Image Enhancement and Image Restoration.

Ans → The main source of noise in an image are as follows:-

- i) Sensor noise
- ii) Environmental noise
- iii) Transmission noise
- iv) Photographic noise

S.N	Image Enhancement	Image Restoration
i)	It is a process which aims to improve bad images so they look better.	It is a process which aims to invert known or estimated degradation to image.
ii)	It provides better visual representation.	It removes effects of sensing environment.
iii)	It is a subjective process.	It is an objective process.
iv)	It is difficult to quantitative measures.	It is easy to quantitative measures by mathematical model.
v)	It is concerned with about extraction of image features.	It is concerned about restoration of degradation.

Image Compression

Introduction:

- Image Compression is a type of data compression applied to digital images, to reduce their cost for storage or transmission.
- Image Compression is a type of data compression techniques that reduces the size of an image file without affecting or degrading its quality to a greater extent.
- Image compression is typically performed through an image/data compression algorithm/codecs.
- Why compression?
 - i) To reduce the volume of data to be transmitted (text, fax, images).
 - ii) To reduce the bandwidth required for transmission and to reduce storage requirements (speech, audio, video).
- How is compression possible?
 - i) Redundancy in digital audio, image and video data.
 - ii) Properties of human perception.
 - iii) Digital audio is a series of sample values; image is a rectangular array of pixel values; video is a sequence of images played out at a certain time.
 - iv) Neighboring sample values are correlated.

Redundancy:

→ There are three basic types of redundancy can be identified in a single image:-

- Coding Redundancy
- Interpixel Redundancy
- Psychovisual Redundancy

5.1. Coding Redundancy:

- Coding redundancy is used to reduce the amount of data representing same information.
- Coding redundancy is always reversible and usually implementing using Look Up Tables (LUTs).
- Let us assume that a discrete random variable r_k in the interval $[0,1]$ represent the gray level of an image that each r_k occurs with probability $P_r(r_k) = \frac{n_k}{n}$ where, $k=0, 1, 2, \dots, L-1$.

where,

L is the no. of gray levels

n_k is the no. of times that the k th gray level appears in the image.

n is the total no. of pixels in the image

- If the no. of bits used to represent each value of r_k is $l(r_k)$ then the average no. of bits, required to represent each pixel is

$$L_{avg} = \sum_{k=0}^{L-1} l(r_k) P_r(r_k)$$

5.1.1. Huffman Coding:

→ The following are the steps of Huffman Coding.

1. Ranking pixel values in decreasing order of their probability.
2. Pair the two values with the lowest probabilities, labelling one of them with 0 and other with 1.
3. Link two symbols with lowest probabilities.
4. Go to step 2 until you generate a single symbol which probability is 1.
5. Trace the coding tree from a root.

Numericals:

① Consider the following source with symbol and their probability. Generate the code word using Huffman coding. Also calculate the compression ratio and efficiency of the system.

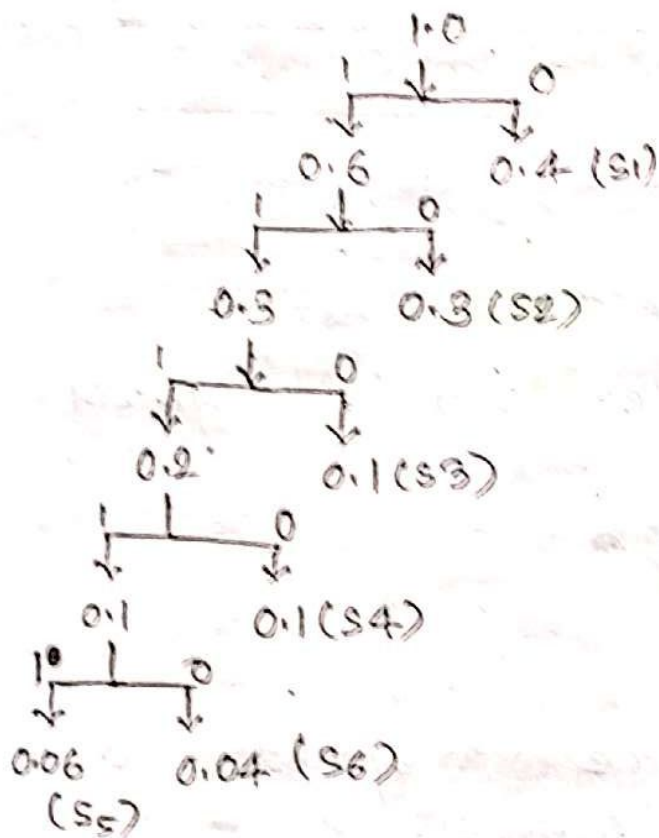
Symbol	S1	S2	S3	S4	S5	S6
Probability	0.1	0.4	0.06	0.1	0.04	0.3

So Here,

Step 1: Source Reduction:

Original Source		Source reduction							
Symbol	Probability	1	2	3	4	5			
S1	0.4	0.4	0.4	0.4	→ 0.6	→ 1.0			
S2	0.3	0.3	0.3	0.3	→ 0.4				
S3	0.1	0.1	→ 0.2	→ 0.3					
S4	0.1	0.1	→ 0.1						
S5	0.06	→ 0.1							
S6	0.04								

Step 2: Construct Huffman tree, assume highest probability $0.6 = 1$ and lowest probability $0.4 = 0$



Step 3: Generate codeword length;

Symbol	Probability (P_i)	Codelength (L_i)
S1	0.4	0 = 1
S2	0.3	10 = 2
S3	0.1	110 = 3
S4	0.1	1110 = 4
S5	0.06	11111 = 5
S6	0.04	11110 = 5

Now

$$\text{Compression ratio, } L_{avg} = \sum_{i=1}^N P_i L_i = \sum_{i=1}^6 P_i L_i$$

$$= P_1 L_1 + P_2 L_2 + P_3 L_3 + P_4 L_4 + P_5 L_5 + P_6 L_6$$

$$= 0.4 \times 1 + 0.3 \times 2 + 0.1 \times 3 + 0.4 \times 4 + 0.06 \times 5 + 0.04 \times 5$$

$$= 2.2 \text{ bits/symbol. } \underline{\text{Ans.}}$$

Also

$$\text{Entropy (H)} = - \sum_{i=1}^N P_i \log_2 P_i = - \sum_{i=1}^6 P_i \log_2 P_i$$

$$= - \left[0.4 \log_2(0.4) + 0.3 \log_2(0.3) + 0.1 \log_2(0.1) + 0.1 \log_2(0.1) + 0.06 \log_2(0.06) + 0.04 \log_2(0.04) \right]$$

$$= 2.14$$

And,

$$\text{Efficiency} = \frac{\text{Entropy (H)}}{L_{avg}} \times 100\% = \frac{2.14}{2.2} \times 100\%$$
$$= 97.27\% \quad \underline{\text{Ans.}}$$

2) Calculate the compression ratio and efficiency from the given image below:

r	0	1	2	3	4	5	6	7
n_k	400	1350	659	2034	516	2560	250	1500

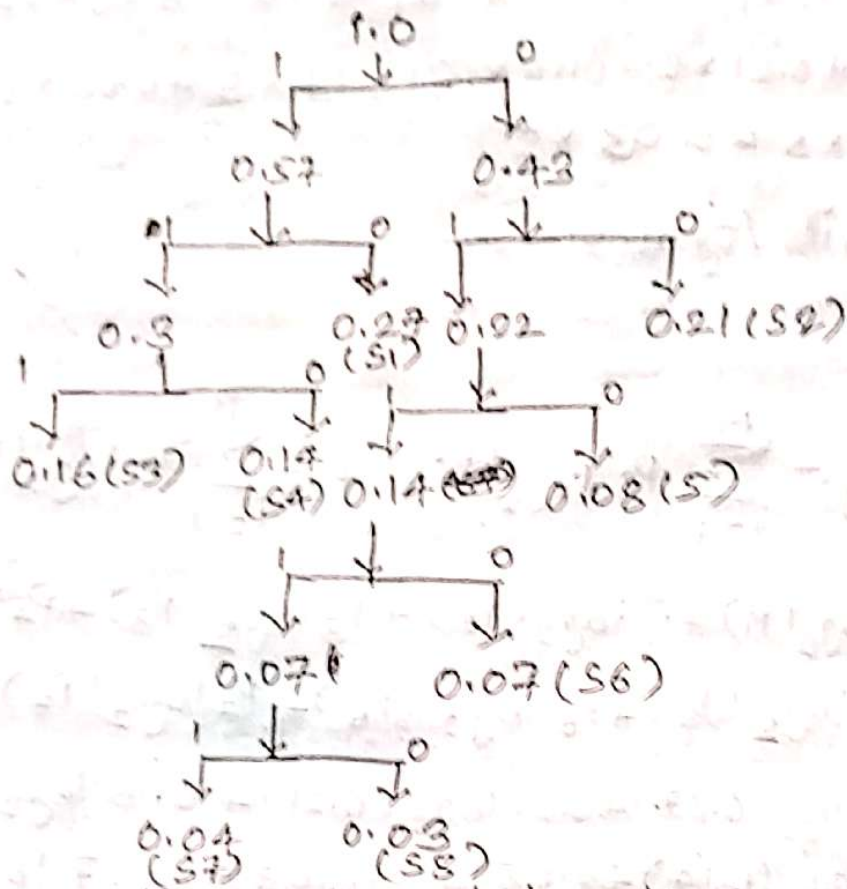
Step 1: calculate the probability in each pixel,

Gray level, r_k	No. of pixel, n_k	Probability (P_i) = n_k/n
0	400	0.04
1	1350	0.14
2	659	0.07
3	2034	0.21
4	516	0.08
5	2560	0.27
6	250	0.03
7	1500	0.16
	$n = 9569$	

Step 2: Source Reduction;

Original Source		Source Reduction						
Symbol	P_i	1	2	3	4	5	6	7
S1	0.27	0.27	0.27	0.27	→ 0.3	→ 0.45	→ 0.57	
S2	0.21	0.21	0.21	→ 0.22	0.27	→ 0.3	→ 0.45	→ 1.0
S3	0.16	0.16	0.16	→ 0.21	0.27	→ 0.27		
S4	0.14	0.14	0.14	→ 0.16	0.21			
S5	0.08	0.08	→ 0.14	→ 0.14				
S6	0.07	0.07	→ 0.08					
S7	0.04							
S8	0.03	→ 0.07						

step 8: Construct Huffman tree, assume highest probability, $0.57 = 1$ and $0.43 = 0$ lowest probability, $0.43 = 0$.



step 4: Generate codeword length,

Symbol	Probability, P_i	Codelength (L_i)
s1	0.27	10 = 2
s2	0.21	00 = 2
s3	0.16	111 = 3
s4	0.14	0110 = 3
s5	0.108	0110 = 3
s6	0.07	0110 = 4
s7	0.04	01111 = 5
s8	0.03	01110 = 5

Ans,
 Compression ratio, $L_{avg} = \sum_{i=1}^N p_i L_i = \sum_{i=1}^8 p_i L_i$

$$= p_1 L_1 + p_2 L_2 + p_3 L_3 + p_4 L_4 + p_5 L_5 + p_6 L_6 + p_7 L_7 + p_8 L_8$$

$$= 0.27 \times 2 + 0.21 \times 2 + 0.16 \times 3 + 0.14 \times 3 + 0.08 \times 3 + 0.07 \times 4$$

$$+ 0.04 \times 5 + 0.03 \times 5$$

$\therefore L_{avg} = 2.73 \text{ bits/symbol}$

Also,

Entropy, $H = - \sum_{i=1}^N p_i \log_2(p_i) = - \sum_{i=1}^8 p_i \log_2(p_i)$

$$= - [p_1 \log_2(p_1) + p_2 \log_2(p_2) + p_3 \log_2(p_3) + p_4 \log_2(p_4)$$

$$+ p_5 \log_2(p_5) + p_6 \log_2(p_6) + p_7 \log_2(p_7) + p_8 \log_2(p_8)]$$

$$= - [0.27 \log_2(0.27) + 0.21 \log_2(0.21) + 0.16 \log_2(0.16) +$$

$$0.14 \log_2(0.14) + 0.08 \log_2(0.08) + 0.07 \log_2(0.07) +$$

$$0.04 \log_2(0.04) + 0.03 \log_2(0.03)]$$

$\therefore H = 2.7$

Again

Efficiency, $\eta = \frac{H}{L_{avg}} \times 100\%$

$$= \frac{2.7}{2.73} \times 100\%$$

$\therefore \eta = 98.9\% \quad \underline{\text{Ans}}$

5.2. Interpixel Redundancy:

- Interpixel redundancy is due to the correlation between the neighboring pixels in the an image.
- That means neighboring pixels are not statistically independent and the gray levels are not equally probable.
- To reduce the inter-pixel redundancy the difference between adjacent pixels can be used to represent an image.

5.2.1. Run Length Coding:

- Run length coding is a lossless compression technique that is used to reduce the size of digital images by replacing repeated data with a count of how many times the data appears.
 - Let us take a hypothetical single scanline with B representing a black pixel and W representing white.
- WWWWWWWWBWWWWBBBWWBBB: original image size
- If we apply the run length coding data compression algorithm to the above hypothetical scanline. We get the following:

7W1B3W3B2W2B: Compressed image size

- The compression ratio (CR):

$$CR = \frac{\text{original image size}}{\text{compressed image size}} = \frac{20}{12} = \frac{5}{3} = 5:3$$

$$\therefore CR = 5:3$$

~~5.3. Phys Psychovisual Redundancy~~

5.2.2. Bit Plane Coding:

- Bit plane coding is a technique that reduces image inter-pixel redundancy by processing each bit plane of an image individually.
- Bit plane coding decomposes an image into a set of binary images, with each bit plane representing a different level of detail.
- Each binary image is then compressed using binary compression methods.
- Let I be the following 2×2 image where the pixels are 3 bits long.

101	110
111	011

- The corresponding 3 bit planes are:

1	1	0	1	1	0
1	0	1	1	1	1

- An m -bit gray scale image can be converted into m -binary images by bit-plane slicing.
- These individual images are then encoded using run-length coding.
- However, a small change in the gray level of adjacent pixels can cause a disruption of the run of zeros or ones.

Eg.

Let one pixel has a gray level of 127 and another pixel has a gray level of 128.

In binary,

$$127 = 01111111 \text{ \&}$$

$$128 = 10000000$$

5.3. Psychovisual Redundancy:

- Psychovisual redundancy is the presence of irrelevant information in an image that can be removed without significantly reducing the image's quality.
- This is because the human eye and brain don't respond to all visual information with the same intensity.
- For example; the human eye is more sensitive to differences in dark intensities than bright ones.
- Psychovisual redundancy can include: artifacts, noise and details that distract from the image's primary content.
- While psychovisually redundant information can be removed, it's important to note that doing so can result in a loss of quantitative information.

~~5.4 Compression Models:~~

5.4. Image Compression Models:

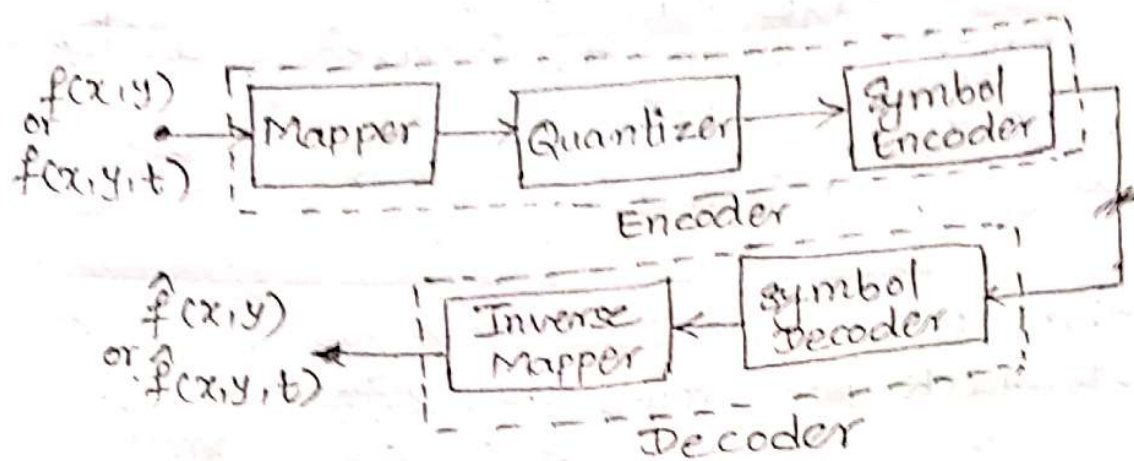


Fig. Image Compression models

→ The image compression models consists of two processes:-

i) Encoder or Encoding Process:

→ The encoder of above figure is designed to remove the redundancies through a series of independent operations.

→ In first stage, the mapper transforms the input data into a format designed to reduce inter-pixel redundancies in the input image. This operation is reversible and may or may not reduce directly the amount of data required to represent the image. Eg. RLE (Run Length Encoding).

→ In second stage, quantizer reduces the accuracy of the mapper's output in accordance with some pre-established fidelity criteria. This stage reduces the psycho-visual redundancies of the input image.

→ In third stage, the symbol encoder creates a fixed or variable length code to represent.

→ In third stage, the symbol encoder creates a fixed or variable length code to represent the quantizer output and maps the output in accordance with the code. In many cases, a variable length code is used encoding is used. This operation is reversible.

Upon its completion, the input image has be processed for the removal of each of the three redundancies.

ii) Decoder or Decoding Process:-

- The decoder of above figure contains only two components: a symbol decoder and an inverse mapper.
- They perform, in reverse order, the inverse operations of the encoder's symbol encoder and mapper.
- Because quantization results in irreversible information loss, an inverse quantizer block is not included in the general decoder model.

5.5. Lossless and Lossy Compression:

SN	Lossy Compression	Lossless Compression
i>	Lossy compression is the method which eliminates the data which is not noticeable.	Lossless compression is the method which does not eliminate the data which is not noticeable.
ii>	In lossy compression, a file does not restore or rebuilt in its original form.	In lossless compression, a file can be restored in its original form.
iii>	Lossy compression compromises data's quality.	Lossless compression does not compromise data's quality.
iv>	Lossy compression reduces the size of data.	Lossless compression does not reduce the size of data.
v>	Lossy compression is used in images, audio, video.	Lossless compression is used in text, images, sound.
vi>	Lossy compression has more data holding capacity.	Lossless compression has less data holding capacity.
vii>	Lossy compression is also known as irreversible compression.	Lossless compression is also known as reversible compression.
viii>	Algorithms used in lossy compression are: Transform coding, Discrete Cosine Transform, Discrete Wavelet Transform, Fractal Compression etc.	Algorithms used in lossless compression are: Run Length Encoding, Lempel-Ziv-Welch, Huffman coding, Arithmetic Encoding etc.

5.5.1. Predictive Coding:

→ Predictive coding is a statistical estimation procedure where future random variables are predicted from past and present observable variables.

→ Image Compression:

i) Lossless Predictive Coding

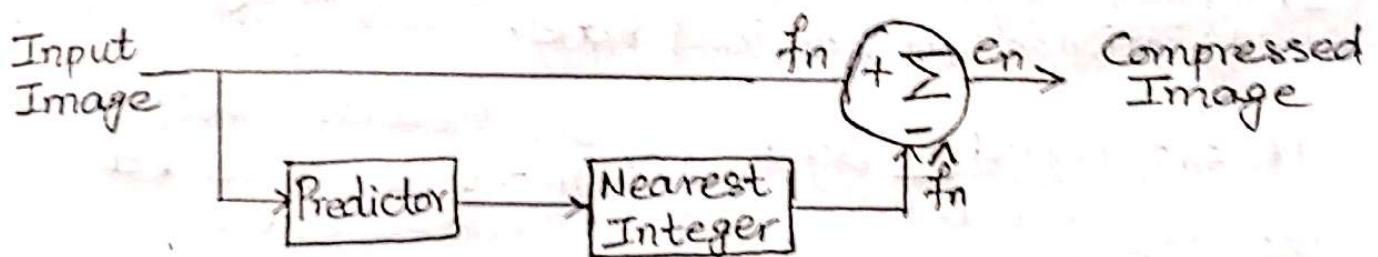
ii) Lossy Predictive Coding

1) Lossless Predictive Coding:

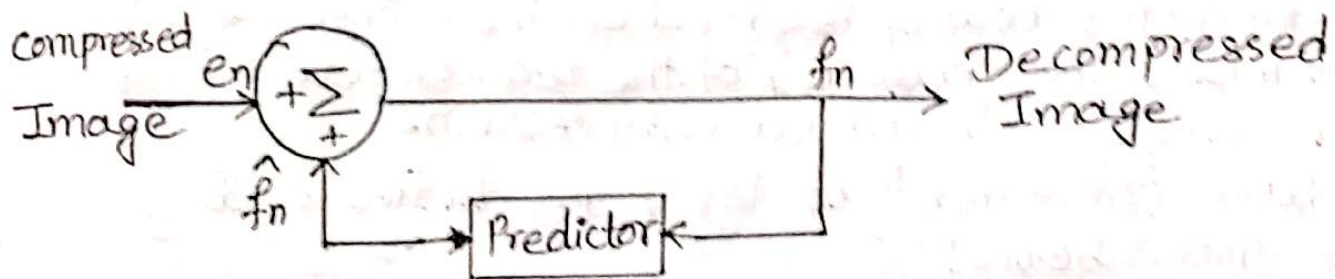
→ lossless predictive coding is based on elimination of inter-pixel redundancy of closely spaced pixels by extracting and coding only the new information.

→ The new information $\Rightarrow$ (Actual pixel Value - Predicted pixel Value)

→ The basic components of lossless predictive coding system are shown below:-



(a) Encoder



(b) Decoder

Fig. Lossless Predictive Coding Model

- The encoder and decoder each contain identical predictor.
- At each pixel successive pixel of input denoted by f_n is introduced to the encoder.
- The predictor generates the anticipated value of that pixel based on some number of past inputs.
- The output of pixel is rounded to the nearest integer denoted by $\hat{f}_n$.
- The prediction error is given by

$$e_n = f_n - \hat{f}_n$$
- In case of encoding, symbol encoder is used.
- In case of decoding, symbol decoder uses decoding function as,

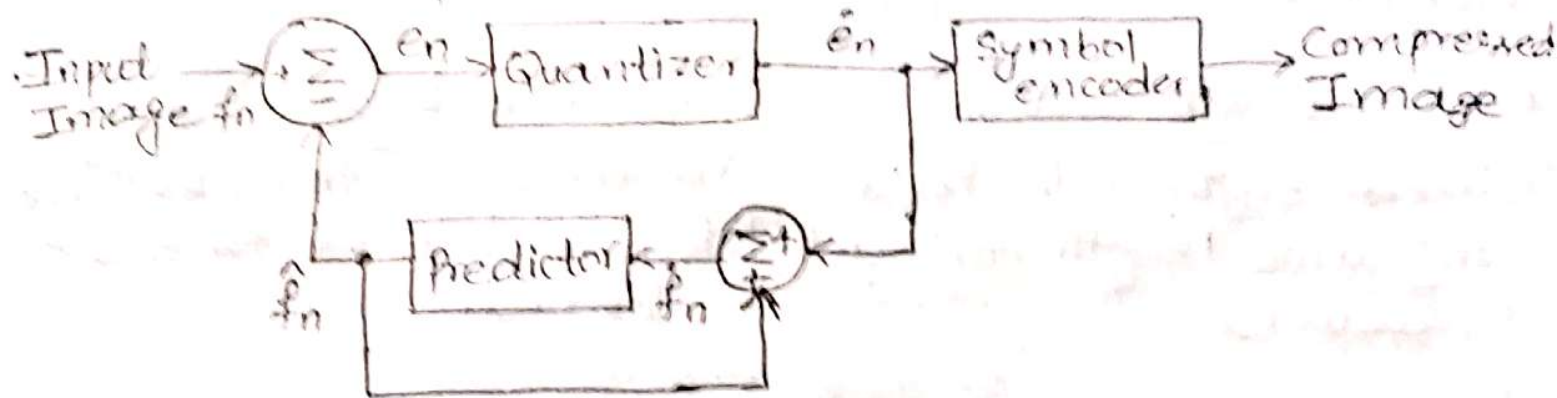
$$f_n = e_n + \hat{f}_n ; \hat{f}_n = \text{decompressed or output image}$$

- In most cases, the predictor is formed by a linear combination of m previous pixels.

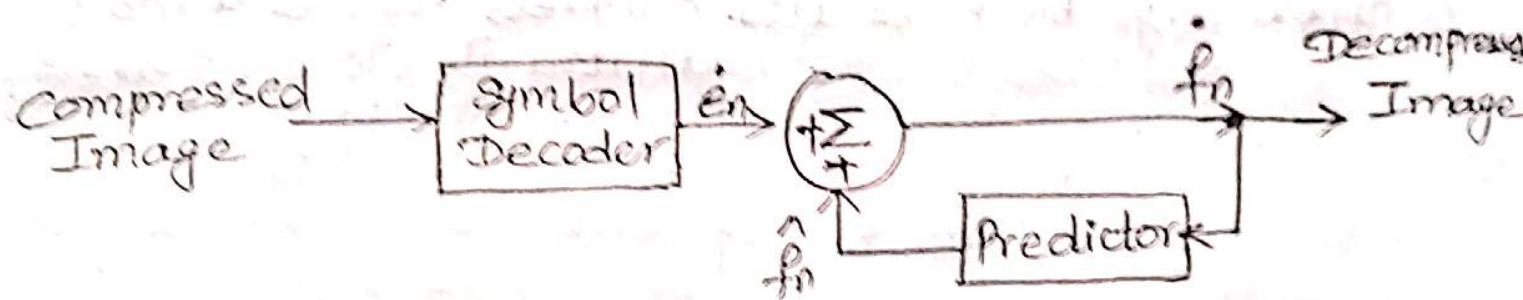
$$\text{i.e. } \hat{f}_n = \text{round} \sum_{i=1}^m \alpha_i f_{n-i} ; \alpha = \text{predictor coefficient} ; m = \text{order of linear prediction}$$

2) Lossy Predictive Coding:

- Lossy predictive coding is based on the concept of compromising the accuracy of the reconstructed image in exchange for increased compression.
- The basic components of lossy predictive coding are shown below:



(a) Encoder



(b) Decoder

Fig. Lossy Predictive Coding Model

- The major difference in this technique is that a quantizer is added between the symbol encoder and the point where the prediction error is calculated.
- The quantizer absorbs the nearest value of a prediction error and maps it into a limited range of output denoted by $\hat{e}_n$ at encoding and $\hat{f}_n$ at decoding, which is defined as;

$$\hat{e}_n = f_n - \hat{f}_n$$

and

$$\hat{f}_n = \hat{e}_n + \hat{f}_n$$

6. Introduction to Morphological Image Processing

- Morphological image processing is a group of non-linear operations that analyze and process an image's geometrical structures and shape.
- Morphological image processing is based on set theory, topology, lattice theory and random functions.
- It is particularly used for processing binary images.
- Its goal is to remove imperfections in an image's structure.

6.1. Logical Operations Involving Binary Images:

- The principal logic operations used in image processing are: AND, OR, NOT (COMPLEMENT). These operations are functionally complete.
- Logical operations are performed on a pixel by pixel basis between corresponding pixels (bitwise).
- Other important logic operations: XOR, NAND.
- Logic operations are just a private case for a binary set operations, such: AND, OR, NOT.

P	q	P AND q	P OR q	NOT P
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

6.2. Erosion and Dilation:

S.N. Erosion	Dilation
i) Erosion removes pixel on object boundaries.	Dilation adds pixels to the boundaries of objects in an image.
ii) Erosion decreases the size of the objects.	Dilation increases the size of the objects.
iii) Erosion removes the small anomalies.	Dilation fills the holes and broken areas.
iv) Erosion decreases the brightness of the bright objects.	Dilation increases the brightness of the objects.
v) Erosion is used later in closing operation.	Dilation is used prior in closing operation.
vi) Erosion eliminates irrelevant detail.	Dilation also preserves pixel detail.
vii) Erosion is denoted as $A \ominus B$.	Dilation is denoted as $A \oplus B$.
viii) For Erosion; - Perfect Match, $PM = 1$ - Some Match, $SM = 0$ - No Match, $NM = 0$	For Dilation; - Perfect Match, $PM = 1$ - Some Match, $SM = 1$ - No Match, $NM = 0$
ix) $A \ominus B = \{z / (B)_z \subseteq A\}$ or $A \ominus B = \{z / (B)_z \cap (A)^c = \emptyset\}$ $\emptyset$ = Empty set B = structuring Element	$A \oplus B = \{z / (\hat{B})_z \cap A \neq \emptyset\}$ $\emptyset$ = Empty set B = structuring Element

Note:

→ Morphological Erosion can be used considered as a noise reduction technique where when applied erosion tends to remove small, isolated noise or small details in the image.

Example:

Find the Erosion and Dilation of the following:

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}_{5 \times 6}; B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}_{3 \times 3}$$

sol:-

$$\text{Erosion} = A \ominus B = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Dilation} = A \oplus B = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

6.3. Opening and closing:

6.3.1. Opening:

- The opening A by B is the erosion of A by B , followed by a dilation of the result by B .
- Opening is denoted by $\circ$.

$$\text{i.e. } A \circ B = (A \ominus B) \oplus B$$

- Erosion followed by dilation is called opening which eliminates protrusions, breaks necks, smoothes contour.

6.3.2. Closing:

- The closing A by B is the erosion dilation of A by B followed by a erosion of the result by B .
- Closing is denoted by $\bullet$.

$$\text{i.e. } A \bullet B = (A \oplus B) \ominus B$$

- Dilation followed by erosion is called closing which smooths contour, eliminates small holes, fuses narrow breaks and long thin gulfs, fills gaps in the contour.

→ Opening and Closing Properties:

- The opening operation satisfies the following properties:
 - A is a subset of $A \circ B$.
 - If C is a subset of D , then $C \circ B$ is a subset of $D \circ B$.

$$(A \circ B) \circ B = A \circ B$$

→ The closing operation properties satisfies the following properties:

- A is a subset of $A \circ B$.
- If C is a subset of D , then $C \circ B$ is a subset of $D \circ B$.

$$(A \circ B) \circ B = A \circ B$$

Example:

① Find the dilation, erosion, opening and closing of set A with structuring element B .

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}_{8 \times 8}$$

$$B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Here,

Dilation: $A \oplus B =$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Erosion: $A \ominus B =$

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	0	1	0	0	0	0	0
0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Now,

Opening: $A \circ B = (A \ominus B) \oplus B =$

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	1	1	0	0	0	0
0	1	1	1	0	0	0	0
0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	0

Closing: $A \bullet B = (A \oplus B) \ominus B =$

0	0	0	0	0	0	0	0
0	0	0	0	0	1	1	0
0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	0
0	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0

7. Image Segmentation

- Image Segmentation is the process of partitioning a digital image into multiple regions (set of pixels).
- The goal of segmentation is to simplify and change the representation of an image into something that is more meaningful and easier to analyze.
- Segmentation accuracy determines the success or failure of computerized analysis procedure.
- Some practical applications of image segmentation are as follows:-
 - i) It is used in medical imaging.
 - ii) It is used to locate object in satellite image.
 - iii) It is used in Face Recognition.
 - iv) It is used in Fingerprint Recognition
 - v) It is used for Machine Vision.
 - vi) It is also used for Iris Recognition

→ Image Segmentation algorithm are generally based on two properties:

- i) Discontinuity
- ii) Similarity (Continuity)

(a) Discontinuity :

- Discontinuity ap is based to partition an image based on abrupt change in intensity such as edge in image.
- There are three basic types of gray level discontinuities in an image i-e, points or spot, lines and edges.

b) Similarity:

→ Similarity is based to partition an image into regions that are similar according to predefined criteria such as thresholding, region growing, region merging and splitting.

7.1. Detection of Discontinuities: